KijaniKiosk Capstone — Project Scope

Track: A — Infrastructure-First | **Repo:** github.com/ob002/kijani-kiosk-capstone | **Author:** Amos Ombworo

Problem statement

Every kk-payments change went straight to the `default` (production) namespace. There was no environment where a release is exercised first, no smoke test, no recorded human approval, and nothing alerted when the service was crash-looping. A bad deploy would be discovered by customers, not by the pipeline. Verifiable at the start: `kubectl get ns` showed no staging namespace and no Jenkins stage deployed or tested anything.

In scope (5)

1. A `kijani-staging` namespace with a ResourceQuota, provisioned by Terraform.

2. A per-environment ConfigMap `kk-payments-config` (different DB_HOST) applied by Ansible to staging and production.

3. One `k8s/deployment.yaml` (probes, resource limits, JSON logs) applied unchanged to both namespaces.

4. A Jenkins pipeline: merge to main triggers it, deploys to staging, runs a smoke test, and offers the production approval gate (reason required) only if the smoke test passes.

5. A Prometheus alert on kk-payments pod restarts, committed in `monitoring/alert-rules.yaml`.

Out of scope

• **Separate production cluster**: no budget for a second cluster; namespace isolation with a quota shows the pattern on one minikube cluster.

• **Alert routing (Alertmanager receivers)**: needs external Slack/email accounts; the requirement is a rule that fires on a real signal.

• **Real database**: DB_HOST separation is the requirement; provisioning a database adds cost without changing the delivery design.

Success criteria (each shown in the live demo)

#	Criterion	How it is demonstrated
S1	Staging namespace exists via Terraform	terraform apply → kubectl get ns kijani-staging shows Active
S2	Configs differ, manifest identical	DB_HOST printed from both namespaces; one deployment.yaml in the repo
S3	Merge to main deploys staging with no clicks	Merge a PR → Jenkins build 'Started by an SCM change' within 2 min
S4	Gate only after smoke PASS; reason recorded	Smoke PASS → approval with reason → reason in build description
S5	Fault handled: no gate, prod untouched	Broken probe → smoke FAIL → staging rolled back, approval skipped
S6	Alert fires on a health signal	KkPaymentsPodRestarting shown as FIRING in Prometheus /alerts

Architecture

See `docs/architecture.png`, submitted alongside. GitHub main → Jenkins (pollSCM) → minikube image build → kijani-staging → smoke test → approval → default; Terraform creates the namespace; Ansible applies the ConfigMaps; Prometheus reads pod restart metrics from both namespaces.